

Supply Chain Analysis

Business Insights Report

1. Revenue & Sales Performance

- **Overall scale:** the business generated total revenue of \$8B from around 25,000 customer orders, at an average selling price of \$256 per order.
- **Balanced across regions:** revenue is well spread across all regions, so the business is not overly dependent on any single market. Latin America generates the highest revenue and the Middle East the lowest, but the gap between regions is small.
- **Balanced across categories and brands:** revenue is also distributed almost evenly across product categories and brands overall, reflecting a well-diversified product and brand portfolio.
- **But brand strength differs by region:** Brand_C leads revenue in Asia-Pacific and Europe, while Brand_E is consistently the weakest performer in Asia-Pacific, Europe, and the Middle East. In the Americas the pattern flips — Brand_A leads in Latin America but is the weakest brand in North America, where Brand_B leads instead.
- **Steady month to month:** monthly revenue stays close to \$1B and remains fairly stable all year, with only small fluctuations rather than sharp seasonal swings.
- **Not festival-dependent:** revenue earned outside festival periods is roughly six times higher than revenue earned during festivals, confirming that steady, year-round demand — not holiday shopping — drives most of the business's growth.
- **Premium purchases happen regularly:** a significant number of orders were priced above their category's average price, showing customers are willing to pay more for certain products — a positive sign for revenue and profit margins.
- **Past performance predicts future demand:** products with strong historical sales tend to keep selling well, and past sales are a reliable indicator of future demand. Higher-priced products also tend to generate more revenue overall.
- **Customer segment affects sales:** statistical testing confirms wholesale and retail customers generate meaningfully different average sales, so customer segment should be factored into sales and marketing planning.

2. Customer Segments (Retail vs. Wholesale)

- Retail and wholesale customers place almost the same number of orders (about 12,500 each), showing both segments contribute equally to overall order volume.
- Website conversion rates are also similar between the two segments, with wholesale customers converting only marginally more often than retail customers.

3. Pricing & Discounts

- **Price is not a differentiator:** the company's average product price is almost identical to competitor pricing across categories. This means the business needs to win on supply chain speed, reliability, and efficiency — not on undercutting competitors on price.

- **Wide price range:** product prices span a broad range rather than clustering around one price point, showing the company caters to customers across different budgets rather than focusing only on low-cost or premium items.
- **Consistent discounting:** average discount rates (around 20%) are nearly identical across all brands, product categories, and customer segments — pointing to one consistent, company-wide discount strategy.
- **Discounts don't drive volume:** average sales are nearly the same across all discount levels, showing that discounting has very little effect on how much customers actually buy.

4. Marketing & Website Performance

- **Investment:** total marketing investment across the business was \$132M.
- **Even spend across categories:** marketing spend is very consistent across all product categories, reflecting an equal-budget approach. Beauty receives the slightly highest average spend and Sports the lowest, though the difference is small.
- **Even traffic across categories:** website traffic is similarly consistent across categories, averaging around 25,000–26,000 visits. Beauty draws the most traffic and Fashion the least, again with only a small gap.
- **More marketing spend ≠ more traffic:** website traffic varies widely even at similar marketing spend levels, showing that budget size alone doesn't reliably drive more visitors — other factors are at play.
- **More traffic ≠ more sales:** higher website traffic doesn't guarantee higher sales. For example, Electronics attracts more visitors than Fashion but actually generates slightly lower average sales, pointing to a gap in how well traffic converts into purchases for some categories.
- **Conversion is steady:** the average website conversion rate is 10.5% and stays consistent both month-to-month and across product categories; statistical testing confirms there is no meaningful difference in conversion rates between categories.
- **Traffic hints at future demand:** higher website traffic is linked to stronger demand in the following period, meaning current traffic can act as an early signal of upcoming demand.

5. Product Demand

- Demand is roughly balanced across all product categories overall. Within that balance, Home products see the highest demand, while Electronics is the least sought-after category.

6. Logistics & Delivery Performance

- **Lead times are consistent:** average delivery lead time is about 18 days, and this stays roughly the same regardless of whether shipments go by Air, Sea, or Ground — no single mode is clearly faster.
- **Delays are a company-wide issue:** only about 23% of deliveries arrive on time; roughly 19,000 orders were delayed versus about 6,000 delivered on time. This pattern holds across every region and every shipping mode, pointing to a systemic bottleneck rather than a problem limited to one area.
- **Air is most reliable, Sea least:** Air freight has the best on-time record and Sea freight the worst, but delays outnumber on-time deliveries across all three shipping modes.
- **Balanced mode usage:** Ground, Air, and Sea shipping are all used in similar volumes, reflecting a balanced transportation strategy rather than heavy reliance on one mode.

- **Latin America handles the most volume:** Latin America records both the highest number of on-time deliveries and the highest number of delayed deliveries, largely because it also handles the highest shipment volume overall.
- **Congestion isn't the main delay driver:** Europe faces the highest port congestion but actually has fewer delays than Latin America, showing that delivery delays are driven by factors beyond congestion alone.
- **Fuel cost isn't the main cost driver either:** Latin America and Europe share the same highest average fuel cost, yet Europe's total logistics costs are higher — confirming other logistics factors matter more than fuel price alone.
- **Weather impact varies slightly by region:** weather disruptions affect all regions in a broadly similar way overall, though Latin America experiences the most operational impact and Europe the least.
- **Delays aren't about spending:** statistical testing shows logistics costs do not significantly differ between on-time and delayed deliveries — delays are not simply a matter of higher or lower logistics spending.
- **Shipping mode does affect cost:** statistical testing shows logistics costs do significantly differ by shipping mode, meaning the choice of shipping method has a real, measurable impact on cost.
- **Cost is consistent by region:** statistical testing shows logistics costs do not significantly differ by region, indicating a fairly balanced, consistent cost structure across the network.
- **Cost hotspots:** the Middle East and North America are the two regions where total logistics costs run above the company's global average.
- **Overall cost and emissions:** total logistics expenditure was \$193M, and total carbon emissions from logistics activity were about 6 million units. Higher shipping costs are linked to higher carbon emissions, and one shipping mode contributes the largest share of total emissions.
- **Lead time affects service quality:** longer delivery lead times are linked to lower service levels and fewer on-time deliveries.
- **Tariffs add to logistics cost:** higher tariff rates are linked to higher total logistics costs.
- **Cost doesn't scale simply with time:** logistics costs vary widely even among deliveries with similar lead times, meaning longer deliveries don't automatically cost more to run.

7. Suppliers & Warehouse Operations

- **Reliable supplier network:** supplier reliability is strong and consistent company-wide, averaging about 85%. North America has the highest reliability and Europe the lowest, though the gap between regions is small.
- **Balanced warehouse network:** warehouse utilization and logistics workload are well balanced across the network, meaning no single warehouse or region is overloaded — a sign of effective capacity planning.
- **Stockout hotspots:** Warehouse 4 experiences the most stockout incidents while Warehouse 3 experiences the fewest. By category, Fashion products face more stockouts than any other category, while Beauty products face the fewest.
- **Overall availability is good:** only about 2.5% of inventory experiences stockouts company-wide (627 incidents in total), indicating generally strong product availability.
- **Inventory stays ahead of demand:** inventory levels remain above forecasted demand throughout the year. The recommended average replenishment quantity is about 50 units, with average inventory levels sitting around 2,600 units.

- **Inventory and utilization move together:** higher inventory levels are linked to higher warehouse utilization; however, warehouses with larger capacity sometimes show lower utilization if that extra capacity isn't fully used.

8. Product Returns

- The average product return rate is about 8% and stays fairly consistent across countries. Among all countries, Japan, Canada, India, the UK, and France show the relatively highest return rates.